1

Introduction to Physics

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Notation

$\mathcal{M}_+(X, \mathcal{A})$	finite measures on $(X, \mathcal{A})$
v	velocity

Preface

cc-by-nc-sa

Chapter 1

Relations

1.1 Binary Heterogeneous Relations

keep this part here, after section on sets similar to analysis notes, so we can assume arbitrary unions, products, etc, not just unions/products/etc of two sets

1.1.1 Introduction

Definition 1.1.1.1. Let A, B be sets.

- Let $R \subset A \times B$. Then R is said to be a **relation over** A **and** B.
- We define $\mathcal{R}(A, B) := \mathcal{P}(A, B)$

Definition 1.1.1.2. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$, $S \in \mathcal{R}(B, C)$. We define the **composition relation of** S **with** R, denoted $S \circ_{\mathbf{Rel}} R \in \mathcal{R}(A, C)$, by

$$S \circ_{\mathbf{Rel}} R := \{(a, c) \in A \times C : \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Note 1.1.1.3. When the context is clear, we write $S \circ R$ in place of $S \circ_{\mathbf{Rel}} R$.

Exercise 1.1.1.4. Let A, B, C, D be sets, R a relation between A and B, S relation between B and C and T relation between C and D. Then $(T \circ S) \circ R = T \circ (S \circ R)$.

Proof.

- Let $(a, d) \in (T \circ S) \circ R$. Then there exists $b \in B$ such that $(a, b) \in R$ and $(b, d) \in T \circ S$. Since $(b, d) \in T \circ S$, there exists $c \in C$ such that $(b, c) \in S$ and $(c, d) \in T$. Since $(a, b) \in R$ and $(b, c) \in S$, $(a, c) \in S \circ R$. Since $(a, c) \in S \circ R$ and $(c, d) \in T$, we have that $(a, d) \in T \circ (S \circ R)$. Since $(a, d) \in (T \circ S) \circ R$ is arbitrary, we have that for each $(a, d) \in (T \circ S) \circ R$, $(a, d) \in T \circ (S \circ R)$. Hence $(T \circ S) \circ R \subset T \circ (S \circ R)$.
- Similarly, $T \circ (S \circ R) \subset (T \circ S) \circ R$.

Since $(T \circ S) \circ R \subset T \circ (S \circ R)$ and $T \circ (S \circ R) \subset (T \circ S) \circ R$, we have that $(T \circ S) \circ R = T \circ (S \circ R)$. □

Definition 1.1.1.5. Let A be a set. We define the **diagonal relation on** A, denoted $\Delta_A \in \mathcal{R}(A, A)$, by

$$\Delta_A := \{(a, a) : a \in A\}.$$

Exercise 1.1.1.6. Let A, B be sets and $R \in \mathcal{R}(A, B)$. Then

1. $R \circ \Delta_A = R$
2. $\Delta_B \circ R = R$

Proof.

1. - Let $(a,b) \in R \circ \Delta_A$. Then there exists $a' \in A$ such that $(a,a') \in \Delta_A$ and $(a',b) \in R$. By definition of Δ_A, $a = a'$. Therefore $(a,b) \in R$. Since $(a,b) \in R \circ \Delta_A$ is arbitrary, we have that for each $(a,b) \in R \circ \Delta_A$, $(a,b) \in R$. Hence $R \circ \Delta_A \subset R$.
   - Let $(a,b) \in R$. Then $a \in A$. Therefore $(a,a) \in \Delta_A$. Since $(a,a) \in \Delta_A$ and $(a,b) \in R$, we have that $(a,b) \in R \circ \Delta_A$. Since $(a,b) \in R$ is arbitrary, we have that for each $(a,b) \in R$, $(a,b) \in R \circ \Delta_A$. Hence $R \circ \Delta_A \subset R$.

   Since $R \circ \Delta_A \subset R$ and $R \circ \Delta_A \subset R$, we have that $R \circ \Delta_A = R$.
2. Similar to (1).

□

Exercise 1.1.1.7. Let A, B, C be sets.

1. Let $R \in \mathcal{R}(A,B)$ and $(S_\gamma)_{\gamma\in\Gamma} \subset \mathcal{R}(B,C)$. Then
   (a) $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right) \circ R = \bigcup_{\gamma\in\Gamma} (S_\gamma \circ R)$,
   (b) $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) \circ R \subset \bigcap_{\gamma\in\Gamma} (S_\gamma \circ R)$
2. Let $(R_\gamma)_{\gamma\in\Gamma} \subset \mathcal{R}(A,B)$ and $S \in \mathcal{R}(B,C)$. Then
   (a) $S \circ \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma} (S \circ R_\gamma)$.
   (b) $S \circ \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \subset \bigcap_{\gamma\in\Gamma} (S \circ R_\gamma)$.

Proof.

1. (a) We note that

$$\begin{aligned}
(a,c) \in \left(\bigcup_{\gamma\in\Gamma} S_\gamma\right) \circ R &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in \bigcup_{\gamma\in\Gamma} S_\gamma \\
&\iff \text{there exists } b \in B \text{ and } \gamma_0 \in \Gamma \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a,c) \in S_{\gamma_0} \circ R \\
&\iff (a,c) \in \bigcup_{\gamma\in\Gamma} (S_\gamma \circ R).
\end{aligned}$$

   Hence $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right) \circ R = \bigcup_{\gamma\in\Gamma} (S_\gamma \circ R)$.

   (b) We note that

$$\begin{aligned}
(a,c) \in \left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) \circ R &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in \bigcap_{\gamma\in\Gamma} S_\gamma \\
&\iff \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and for each } \gamma \in \Gamma \text{ and } (b,c) \in S_\gamma \\
&\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (a,c) \in S_\gamma \circ R \\
&\iff (a,c) \in \bigcap_{\gamma\in\Gamma} (S_\gamma \circ R).
\end{aligned}$$

   Hence $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) \circ R = \bigcap_{\gamma\in\Gamma} (S_\gamma \circ R)$.

2. (a) We note that

$$\begin{aligned}(a,c) \in S \circ \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) &\iff \text{there exists } b \in B \text{ such that } (a,b) \in \bigcup_{\gamma\in\Gamma} R_\gamma \text{ and } (b,c) \in S \\ &\iff \text{there exists } b \in B \text{ and } \gamma_0 \in \Gamma \text{ such that } (a,b) \in R_{\gamma_0} \text{ and } (b,c) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a,c) \in S \circ R_{\gamma_0} \\ &\iff (a,c) \in \bigcup_{\gamma\in\Gamma} (S \circ R_\gamma).\end{aligned}$$

Hence $S \circ \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma} (S_\gamma \circ R)$.

(b) We note that

$$\begin{aligned}(a,c) \in S \circ \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) &\iff \text{there exists } b \in B \text{ such that } (a,b) \in \bigcap_{\gamma\in\Gamma} R_\gamma \text{ and } (b,c) \in S \\ &\iff \text{there exists } b \in B \text{ such that for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (b,c) \in S \\ &\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a,b) \in R_\gamma \text{ and } (b,c) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ (a,c) \in S \circ R_\gamma \\ &\iff (a,c) \in \bigcap_{\gamma\in\Gamma} (S \circ R_\gamma).\end{aligned}$$

Hence $S \circ \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \subset \bigcap_{\gamma\in\Gamma} (S_\gamma \circ R)$.

□

Exercise 1.1.1.8. Let A, B, C be sets.

1. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$, then $S \circ R_1 \subset S \circ R_2$.
2. Let $R \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $S_1 \subset S_2$, then $S_1 \circ R \subset S_2 \circ R$.
3. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$ and $S_1 \subset S_2$, then $S_1 \circ R_1 \subset S_2 \circ R_2$.

Proof.

1. Suppose that $R_1 \subset R_2$. Then

$$\begin{aligned}(a,c) \in S \circ R_1 &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R_1 \text{ and } (b,c) \in S \\ &\implies \text{there exists } b \in B \text{ such that } (a,b) \in R_2 \text{ and } (b,c) \in S \\ &\iff (a,c) \in S \circ R_2.\end{aligned}$$

Hence $S \circ R_1 \subset S \circ R_2$.

2. Suppose that $S_1 \subset S_2$. Then

$$\begin{aligned}(a,c) \in S_1 \circ R &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_1 \\ &\implies \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_2 \\ &\iff (a,c) \in S_2 \circ R.\end{aligned}$$

Hence $S_2 \circ R \subset S_1 \circ R$.

3. Suppose that $R_1 \subset R_2$ and $S_1 \subset S_2$. Then the previous parts imply that

$$\begin{aligned}S_1 \circ R_1 &\subset S_2 \circ R_1 \\ &\subset S_2 \circ R_2.\end{aligned}$$

□

Definition 1.1.1.9. Let A, B be sets and $R \in \mathcal{R}(A, B)$. We define the **converse of** R, denoted $R^* \in \mathcal{R}(B, A)$, by

$$R^* := \{(b, a) \in B \times A : (a, b) \in R\}.$$

Exercise 1.1.1.10. Let A, B be sets and R a relation between A and B. Then $(R^*)^* = R$.

Proof.

$$\begin{aligned} (a, b) \in (R^*)^* &\iff (b, a) \in R^* \\ &\iff (a, b) \in R. \end{aligned}$$

Hence $(R^*)^* = R$. □

Exercise 1.1.1.11. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(B, C)$. Then $(S \circ R)^* = R^* \circ S^*$.

Proof. We note that

$$\begin{aligned} (c, a) \in (S \circ R)^* &\iff (a, c) \in S \circ R \\ &\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S \\ &\iff \text{there exists } b \in B \text{ such that } (c, b) \in S^* \text{ and } (b, a) \in R^* \\ &\iff (c, a) \in R^* \circ S^*. \end{aligned}$$

Hence $(S \circ R)^* = R^* \circ S^*$. □

Exercise 1.1.1.12. Let A and B be sets and $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$. Then

1. $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$
2. $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$

Proof.

1. We note that

$$\begin{aligned} (b, a) \in \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (b, a) \in R_{\gamma_0}^* \\ &\iff (b, a) \in \bigcup_{\gamma \in \Gamma} R_\gamma^*. \end{aligned}$$

   Hence $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$.

2. We note that

$$\begin{aligned} (b, a) \in \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \\ &\iff \text{for each } \gamma \in \Gamma,\ (a, b) \in R_\gamma \\ &\iff \text{for each } \gamma \in \Gamma,\ (b, a) \in R_\gamma^* \\ &\iff (b, a) \in \bigcap_{\gamma \in \Gamma} R_\gamma^*. \end{aligned}$$

   Hence $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$.

□

1.1.2 Subrelations

Definition 1.1.2.1. Let $n \in \mathbb{N}$, A, B, E, F sets, $R \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(E, F)$. Then

- S is said to be a **restriction of** R if $E \subset A$ and $F \subset B$,
- R is said to be an **extension of** S if S is a restriction of R.

Definition 1.1.2.2. Let A, B be sets, $E \subset A$, $F \subset B$ and $R \in \mathcal{R}(A, B)$. We define the **restriction of** R **over** E **and** F, denoted $R|_{E\times F} \in \mathcal{R}(E, F)$, by $R|_{E\times F} := R \cap (E \times F)$.

Exercise 1.1.2.3. Let A, B be sets, $E \subset A$, $F \subset B$ and $R \in \mathcal{R}(A, B)$. Then $R|_B$ is a restriction of R.

Proof. Clear. fill out later □

maybe dont do this subsection, since relations are like morphisms, so unless you can find a categorical way to present restrictions on domain or codomain $f|_E$ or $f|^F$, just get rid of this subsection

Exercise 1.1.2.4. Let A, B be sets, $E \subset A$, $F \subset B$ and $R \in \mathcal{R}(A, B)$. Suppose that $E \subset A$ and $F \subset B$. Then $(R^*)|_{F\times E} = (R|_{E\times F})^*$.

Proof. We note that for each $b, a \in B \times A$,

$$\begin{aligned}(b, a) \in (R^*)|_{F\times E} &\iff (b, a) \in R^* \cap (F \times E)\\ &\iff (a, b) \in R \cap (E \times F)\\ &\iff (a, b) \in R|_{E\times F}.\end{aligned}$$

Hence $(R^*)|_{F\times E} = R|_{E\times F}$. □

1.1.3 Products of Relations

Definition 1.1.3.1. Let A, B, C, D be sets and $R \in \mathcal{R}(A, B), S \in \mathcal{R}(C, D)$. We define the **product relation of** R **and** S, denoted $R \times_{\mathbf{Rel}} S \in \mathcal{R}(A \times C, B \times D)$, by

$$R \times_{\mathbf{Rel}} S := \{((a, c), (b, d)) \in (A \times C) \times (B \times D) : (a, b) \in R \text{ and } (c, d) \in S\}.$$

Note 1.1.3.2. When the context is clear, we write $R \times S$ in place of $R \times_{\mathbf{Rel}} S$.

Exercise 1.1.3.3. Let A, B, C, D be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma\in\Gamma} \subset \mathcal{R}(C, D)$. Then
   (a) $R \times \left(\bigcup_{\gamma\in\Gamma} S_\gamma\right) = \bigcup_{\gamma\in\Gamma} (R \times S_\gamma)$
   (b) $R \times \left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) = \bigcap_{\gamma\in\Gamma} (R \times S_\gamma)$
2. Let $(R_\gamma)_{\gamma\in\Gamma} \subset\in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(C, D)$. Then
   (a) $\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcup_{\gamma\in\Gamma} (R_\gamma \times S)$
   (b) $\left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcap_{\gamma\in\Gamma} (R_\gamma \times S)$

Proof.

1. (a) We note that

$$\begin{aligned}((a,c),(b,d)) \in R \times \left(\bigcup_{\gamma\in\Gamma} S_\gamma\right) &\iff (a,b) \in R \text{ and } (c,d) \in \bigcup_{\gamma\in\Gamma} S_\gamma \\ &\iff (a,b) \in R \text{ and there exists } \gamma_0 \in \Gamma \text{ such that } (c,d) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a,b) \in R \text{ and } (c,d) \in S_{\gamma_0} \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a,c),(b,d)) \in R \times S_{\gamma_0} \\ &\iff ((a,c),(b,d)) \in \bigcup_{\gamma\in\Gamma}(R \times S_\gamma).\end{aligned}$$

Hence $R \times \left(\bigcup_{\gamma\in\Gamma} S_\gamma\right) = \bigcup_{\gamma\in\Gamma}(R \times S_\gamma)$.

(b) We note that

$$\begin{aligned}((a,c),(b,d)) \in R \times \left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) &\iff (a,b) \in R \text{ and } (c,d) \in \bigcap_{\gamma\in\Gamma} S_\gamma \\ &\iff (a,b) \in R \text{ for each } \gamma \in \Gamma,\ (c,d) \in S_\gamma \\ &\iff \text{for each } \gamma \in \Gamma,\ (a,b) \in R \text{ and } (c,d) \in S_\gamma \\ &\iff \text{for each } \gamma \in \Gamma,\ ((a,c),(b,d)) \in R \times S_\gamma \\ &\iff ((a,c),(b,d)) \in \bigcap_{\gamma\in\Gamma}(R \times S_\gamma).\end{aligned}$$

Hence $R \times \left(\bigcap_{\gamma\in\Gamma} S_\gamma\right) = \bigcap_{\gamma\in\Gamma}(R \times S_\gamma)$.

2. (a) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a,b) \in R_{\gamma_0} \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a,c),(b,d)) \in R_{\gamma_0} \times S \\ &\iff ((a,c),(b,d)) \in \bigcup_{\gamma\in\Gamma}(R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcup_{\gamma\in\Gamma}(R_\gamma \times S)$.

(b) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ ((a,c),(b,d)) \in R_\gamma \times S \\ &\iff ((a,c),(b,d)) \in \bigcap_{\gamma\in\Gamma}(R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcap_{\gamma\in\Gamma}(R_\gamma \times S)$.

$\square$

1.2 n-ary Heterogeneous Relations

1.2.1 Introduction

can we define composition in an operadic way?

Definition 1.2.1.1. Let $n \in \mathbb{N}$ and $(A_j)_{j\in[n]}$ a collection of sets.

- Let $R \subset \prod_{j\in[n]} A_j$. Then R is said to be a **n-ary relation over** $(A_j)_{j\in[n]}$.
- We define $\mathcal{R}(A_j : j \in [n]) := \mathcal{P}(\prod_{j\in[n]} A_j)$

Definition 1.2.1.2. FIX rest of section!!!! (right now it is same as binary relations section) Let A, B, C be sets, $R \in \mathcal{R}(A, B)$, $S \in \mathcal{R}(B, C)$. We define the **composition relation of S with R**, denoted $S \circ_{\mathbf{Rel}} R \in \mathcal{R}(A, C)$, by

$$S \circ_{\mathbf{Rel}} R := \{(a, c) \in A \times C : \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S\}.$$

Note 1.2.1.3. When the context is clear, we write $S \circ R$ in place of $S \circ_{\mathbf{Rel}} R$.

Exercise 1.2.1.4. Let A, B, C, D be sets, R a relation between A and B, S relation between B and C and T relation between C and D. Then $(T \circ S) \circ R = T \circ (S \circ R)$.

Proof.

- Let $(a, d) \in (T \circ S) \circ R$. Then there exists $b \in B$ such that $(a, b) \in R$ and $(b, d) \in T \circ S$. Since $(b, d) \in T \circ S$, there exists $c \in C$ such that $(b, c) \in S$ and $(c, d) \in T$. Since $(a, b) \in R$ and $(b, c) \in S$, $(a, c) \in S \circ R$. Since $(a, c) \in S \circ R$ and $(c, d) \in T$, we have that $(a, d) \in T \circ (S \circ R)$. Since $(a, d) \in (T \circ S) \circ R$ is arbitrary, we have that for each $(a, d) \in (T \circ S) \circ R$, $(a, d) \in T \circ (S \circ R)$. Hence $(T \circ S) \circ R \subset T \circ (S \circ R)$.
- Similarly, $T \circ (S \circ R) \subset (T \circ S) \circ R$.

Since $(T \circ S) \circ R \subset T \circ (S \circ R)$ and $T \circ (S \circ R) \subset (T \circ S) \circ R$, we have that $(T \circ S) \circ R = T \circ (S \circ R)$. □

Definition 1.2.1.5. Let A be a set. We define the **diagonal relation on** A, denoted $\Delta_A \in \mathcal{R}(A, A)$, by

$$\Delta_A := \{(a, a) : a \in A\}.$$

Exercise 1.2.1.6. Let A, B be sets and $R \in \mathcal{R}(A, B)$. Then

1. $R \circ \Delta_A = R$
2. $\Delta_B \circ R = R$

Proof.

1.
   - Let $(a, b) \in R \circ \Delta_A$. Then there exists $a' \in A$ such that $(a, a') \in \Delta_A$ and $(a', b) \in R$. By definition of Δ_A, $a = a'$. Therefore $(a, b) \in R$. Since $(a, b) \in R \circ \Delta_A$ is arbitrary, we have that for each $(a, b) \in R \circ \Delta_A$, $(a, b) \in R$. Hence $R \circ \Delta_A \subset R$.
   - Let $(a, b) \in R$. Then $a \in A$. Therefore $(a, a) \in \Delta_A$. Since $(a, a) \in \Delta_A$ and $(a, b) \in R$, we have that $(a, b) \in R \circ \Delta_A$. Since $(a, b) \in R$ is arbitrary, we have that for each $(a, b) \in R$, $(a, b) \in R \circ \Delta_A$. Hence $R \circ \Delta_A \subset R$.

   Since $R \circ \Delta_A \subset R$ and $R \circ \Delta_A \subset R$, we have that $R \circ \Delta_A = R$.
2. Similar to (1).

□

Exercise 1.2.1.7. Let A, B, C be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma\in\Gamma} \subset \mathcal{R}(B, C)$. Then
   (a) $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right) \circ R = \bigcup_{\gamma\in\Gamma} (S_\gamma \circ R)$,

(b) $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R \subset \bigcap_{\gamma\in\Gamma}(S_\gamma \circ R)$

2. Let $(R_\gamma)_{\gamma\in\Gamma} \subset \mathcal{R}(A,B)$ and $S \in \mathcal{R}(B,C)$. Then

   (a) $S \circ \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma}(S\circ R_\gamma)$.

   (b) $S \circ \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \subset \bigcap_{\gamma\in\Gamma}(S\circ R_\gamma)$.

Proof.

1. (a) We note that

$$\begin{aligned}
(a,c) \in \left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R &\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in \bigcup_{\gamma\in\Gamma} S_\gamma \\
&\iff \text{there exists } b\in B \text{ and } \gamma_0\in\Gamma \text{ such that } (a,b)\in R \text{ and } (b,c)\in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0\in\Gamma \text{ such that } (a,c)\in S_{\gamma_0}\circ R \\
&\iff (a,c)\in \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R).
\end{aligned}$$

   Hence $\left(\bigcup_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$.

   (b) We note that

$$\begin{aligned}
(a,c) \in \left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R &\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in \bigcap_{\gamma\in\Gamma} S_\gamma \\
&\iff \text{there exists } b\in B \text{ such that } (a,b)\in R \text{ and for each } \gamma\in\Gamma \text{ and } (b,c)\in S_\gamma \\
&\implies \text{for each } \gamma\in\Gamma, \text{ there exists } b\in B \text{ such that } (a,b)\in R \text{ and } (b,c)\in S_\gamma \\
&\iff \text{for each } \gamma\in\Gamma,\ (a,c)\in S_\gamma\circ R \\
&\iff (a,c)\in \bigcap_{\gamma\in\Gamma}(S_\gamma\circ R).
\end{aligned}$$

   Hence $\left(\bigcap_{\gamma\in\Gamma} S_\gamma\right)\circ R = \bigcap_{\gamma\in\Gamma}(S_\gamma\circ R)$.

2. (a) We note that

$$\begin{aligned}
(a,c) \in S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) &\iff \text{there exists } b\in B \text{ such that } (a,b)\in \bigcup_{\gamma\in\Gamma} R_\gamma \text{ and } (b,c)\in S \\
&\iff \text{there exists } b\in B \text{ and } \gamma_0\in\Gamma \text{ such that } (a,b)\in R_{\gamma_0} \text{ and } (b,c)\in S \\
&\iff \text{there exists } \gamma_0\in\Gamma \text{ such that } (a,c)\in S\circ R_{\gamma_0} \\
&\iff (a,c)\in \bigcup_{\gamma\in\Gamma}(S\circ R_\gamma).
\end{aligned}$$

   Hence $S\circ\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) = \bigcup_{\gamma\in\Gamma}(S_\gamma\circ R)$.

(b) We note that

$$\begin{aligned}
(a,c) \in S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) &\iff \text{there exists } b \in B \text{ such that } (a,b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \text{ and } (b,c) \in S \\
&\iff \text{there exists } b \in B \text{ such that for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (b,c) \in S \\
&\implies \text{for each } \gamma \in \Gamma, \text{ there exists } b \in B \text{ such that } (a,b) \in R_\gamma \text{ and } (b,c) \in S \\
&\iff \text{for each } \gamma \in \Gamma,\ (a,c) \in S \circ R_\gamma \\
&\iff (a,c) \in \bigcap_{\gamma \in \Gamma} (S \circ R_\gamma).
\end{aligned}$$

Hence $S \circ \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \subset \bigcap_{\gamma \in \Gamma} (S_\gamma \circ R)$.

□

Exercise 1.2.1.8. Let A, B, C be sets.

1. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$, then $S \circ R_1 \subset S \circ R_2$.
2. Let $R \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $S_1 \subset S_2$, then $S_1 \circ R \subset S_2 \circ R$.
3. Let $R_1, R_2 \in \mathcal{R}(A,B)$ and $S_1, S_2 \in \mathcal{R}(B,C)$. If $R_1 \subset R_2$ and $S_1 \subset S_2$, then $S_1 \circ R_1 \subset S_2 \circ R_2$.

Proof.

1. Suppose that $R_1 \subset R_2$. Then

$$\begin{aligned}
(a,c) \in S \circ R_1 &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R_1 \text{ and } (b,c) \in S \\
&\implies \text{there exists } b \in B \text{ such that } (a,b) \in R_2 \text{ and } (b,c) \in S \\
&\iff (a,c) \in S \circ R_2.
\end{aligned}$$

   Hence $S \circ R_1 \subset S \circ R_2$.

2. Suppose that $S_1 \subset S_2$. Then

$$\begin{aligned}
(a,c) \in S_1 \circ R &\iff \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_1 \\
&\implies \text{there exists } b \in B \text{ such that } (a,b) \in R \text{ and } (b,c) \in S_2 \\
&\iff (a,c) \in S_2 \circ R.
\end{aligned}$$

   Hence $S_2 \circ R \subset S_1 \circ R$.

3. Suppose that $R_1 \subset R_2$ and $S_1 \subset S_2$. Then the previous parts imply that

$$\begin{aligned}
S_1 \circ R_1 &\subset S_2 \circ R_1 \\
&\subset S_2 \circ R_2.
\end{aligned}$$

□

Definition 1.2.1.9. Let A, B be sets and $R \in \mathcal{R}(A,B)$. We define the **converse of** R, denoted $R^* \in \mathcal{R}(B,A)$, by

$$R^* := \{(b,a) \in B \times A : (a,b) \in R\}.$$

Exercise 1.2.1.10. Let A, B be sets and R a relation between A and B. Then $(R^*)^* = R$.

Proof.

$$\begin{aligned}
(a,b) \in (R^*)^* &\iff (b,a) \in R^* \\
&\iff (a,b) \in R.
\end{aligned}$$

Hence $(R^*)^* = R$. □

Exercise 1.2.1.11. Let A, B, C be sets, $R \in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(B, C)$. Then $(S \circ R)^* = R^* \circ S^*$.

Proof. We note that

$$\begin{aligned}
(c, a) \in (S \circ R)^* &\iff (a, c) \in S \circ R \\
&\iff \text{there exists } b \in B \text{ such that } (a, b) \in R \text{ and } (b, c) \in S \\
&\iff \text{there exists } b \in B \text{ such that } (c, b) \in S^* \text{ and } (b, a) \in R^* \\
&\iff (c, a) \in R^* \circ S^*.
\end{aligned}$$

Hence $(S \circ R)^* = R^* \circ S^*$. □

Exercise 1.2.1.12. Let A and B be sets and $(R_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(A, B)$. Then

1. $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$
2. $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$

Proof.

1. We note that

$$\begin{aligned}
(b, a) \in \left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcup_{\gamma \in \Gamma} R_\gamma \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (b, a) \in R_{\gamma_0}^* \\
&\iff (b, a) \in \bigcup_{\gamma \in \Gamma} R_\gamma^*.
\end{aligned}$$

   Hence $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcup_{\gamma \in \Gamma} R_\gamma^*$.

2. We note that

$$\begin{aligned}
(b, a) \in \left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* &\iff (a, b) \in \bigcap_{\gamma \in \Gamma} R_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (a, b) \in R_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (b, a) \in R_\gamma^* \\
&\iff (b, a) \in \bigcap_{\gamma \in \Gamma} R_\gamma^*.
\end{aligned}$$

   Hence $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right)^* = \bigcap_{\gamma \in \Gamma} R_\gamma^*$.

□

1.2.2 Subrelations

Definition 1.2.2.1. Let A, B be sets, $R \in \mathcal{R}(A, B)$, $E \subset A$ and $F \subset B$. We define the **subrelation of R between E and F**, denoted $R|_{E \times F} \in \mathcal{R}(E, F)$, by $R|_{E \times F}$

maybe dont do this subsection, since relations are like morphisms, so unless you can find a categorical way to present restrictions on domain or codomain $f|_E$ or $f|^F$, just get rid of this subsection

1.2.3 Products of Relations

Definition 1.2.3.1. Let A, B, C, D be sets and $R \in \mathcal{R}(A, B), S \in \mathcal{R}(C, D)$. We define the **product relation of R and S**, denoted $R \times_{\mathbf{Rel}} S \in \mathcal{R}(A \times C, B \times D)$, by

$$R \times_{\mathbf{Rel}} S := \{((a, c), (b, d)) \in (A \times C) \times (B \times D) : (a, b) \in R \text{ and } (c, d) \in S\}.$$

Note 1.2.3.2. When the context is clear, we write $R \times S$ in place of $R \times_{\mathbf{Rel}} S$.

Exercise 1.2.3.3. Let A, B, C, D be sets.

1. Let $R \in \mathcal{R}(A, B)$ and $(S_\gamma)_{\gamma \in \Gamma} \subset \mathcal{R}(C, D)$. Then
   (a) $R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$
   (b) $R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$
2. Let $(R_\gamma)_{\gamma \in \Gamma} \subset\in \mathcal{R}(A, B)$ and $S \in \mathcal{R}(C, D)$. Then
   (a) $\left(\bigcup_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcup_{\gamma \in \Gamma} (R_\gamma \times S)$
   (b) $\left(\bigcap_{\gamma \in \Gamma} R_\gamma \right) \times S = \bigcap_{\gamma \in \Gamma} (R_\gamma \times S)$

Proof.

1. (a) We note that

$$\begin{aligned}
((a, c), (b, d)) \in R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcup_{\gamma \in \Gamma} S_\gamma \\
&\iff (a, b) \in R \text{ and there exists } \gamma_0 \in \Gamma \text{ such that } (c, d) \in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a, b) \in R \text{ and } (c, d) \in S_{\gamma_0} \\
&\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a, c), (b, d)) \in R \times S_{\gamma_0} \\
&\iff ((a, c), (b, d)) \in \bigcup_{\gamma \in \Gamma} (R \times S_\gamma).
\end{aligned}$$

   Hence $R \times \left(\bigcup_{\gamma \in \Gamma} S_\gamma \right) = \bigcup_{\gamma \in \Gamma} (R \times S_\gamma)$.

   (b) We note that

$$\begin{aligned}
((a, c), (b, d)) \in R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) &\iff (a, b) \in R \text{ and } (c, d) \in \bigcap_{\gamma \in \Gamma} S_\gamma \\
&\iff (a, b) \in R \text{ for each } \gamma \in \Gamma,\ (c, d) \in S_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ (a, b) \in R \text{ and } (c, d) \in S_\gamma \\
&\iff \text{for each } \gamma \in \Gamma,\ ((a, c), (b, d)) \in R \times S_\gamma \\
&\iff ((a, c), (b, d)) \in \bigcap_{\gamma \in \Gamma} (R \times S_\gamma).
\end{aligned}$$

   Hence $R \times \left(\bigcap_{\gamma \in \Gamma} S_\gamma \right) = \bigcap_{\gamma \in \Gamma} (R \times S_\gamma)$.

2. (a) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } (a,b) \in R_{\gamma_0} \text{ and } (c,d) \in S \\ &\iff \text{there exists } \gamma_0 \in \Gamma \text{ such that } ((a,c),(b,d)) \in R_{\gamma_0} \times S \\ &\iff ((a,c),(b,d)) \in \bigcup_{\gamma\in\Gamma} (R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcup_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcup_{\gamma\in\Gamma} (R_\gamma \times S)$.

(b) We note that

$$\begin{aligned}((a,c),(b,d)) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S &\iff (a,b) \in \left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ (a,b) \in R_\gamma \text{ and } (c,d) \in S \\ &\iff \text{for each } \gamma \in \Gamma,\ ((a,c),(b,d)) \in R_\gamma \times S \\ &\iff ((a,c),(b,d)) \in \bigcap_{\gamma\in\Gamma} (R_\gamma \times S)\end{aligned}$$

Hence $\left(\bigcap_{\gamma\in\Gamma} R_\gamma\right) \times S = \bigcap_{\gamma\in\Gamma} (R_\gamma \times S)$.

$\square$

1.3 n-ary Homogeneous Relations

1.3.1 Introduction

Definition 1.3.1.1. Let $n \in \mathbb{N}$ and A a set.

- Let $R \subset A^n$. Then R is said to be a **homogeneous n-ary relation on** A
- We define $\mathcal{R}^n(A) := \mathcal{P}(A^n)$.

1.3.2 Subrelations

Definition 1.3.2.1. Let $n \in \mathbb{N}$, A, B sets, $R \in \mathcal{R}^n(A)$ and $S \in \mathcal{R}^n(B)$. Then

- S is said to be a **restriction of** R if $B \subset A$,
- R is said to be an **extension of** S if S is a restriction of R.

Definition 1.3.2.2. Let $n \in \mathbb{N}$, A a set, $B \subset A$ and $R \in \mathcal{R}^n(A)$. We define the **restriction of** R **to** B, denoted $R|_B$, by $R|_B := R \cap B^n$.

Exercise 1.3.2.3. Let $n \in \mathbb{N}$, A a set, $B \subset A$ and $R \in \mathcal{R}^n(A)$. Then $R|_B$ is a restriction of R.

Proof. Clear. fill out later □

1.3.3 Product Relations

1.4 Binary Homogeneous Relations

1.4.1 Subrelations

Exercise 1.4.1.1. Let A be a set, $B \subset A$ and $R \in \mathcal{R}^2(A)$. Then $(R^*)|_B = (R|_B)^*$

Proof. Clear by prev ex in section on binary hetero relations. $\square$

1.5 Equivalence Relations

Note 1.5.0.1. Let X be a set and $\nu \in \mathcal{R}^2(X)$. When the context is clear, we may write $a\nu b$ in place of $(a, b) \in \nu$.

Definition 1.5.0.2. Let X be a set. Suppose that $X \neq \varnothing$.

- Let $\theta \in \mathcal{R}^2(X)$. Then θ is said to be
  - **reflexive** if for each $a \in X$, $(a, a) \in \theta$
  - **symmetric** if for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$
  - **transitive** if for each $a, b, c \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$
  - an **equivalence relation on** X if θ is reflexive, symmetric and transitive
- We define $\mathrm{Equiv}(X) := \{\theta \in \mathcal{R}^2(X) : \theta \text{ is an equivalence relation on } X\}$

Exercise 1.5.0.3. Let X be a set and $\theta \in \mathcal{R}^2(X)$. Then

1. θ is reflexive iff $\Delta_X \subset \theta$
2. θ is symmetric iff $\theta^* \subset \theta$
3. θ is transitive iff $\theta \circ \theta \subset \theta$

Proof.

1.
   - $(\implies)$:
     Suppose that θ is reflexive. Let $(a, a) \in \Delta_X$. Then $a \in X$. Since θ is reflexive, $(a, a) \in \theta$. Since $(a, a) \in \Delta_X$ is arbitrary, we have that for each $(a, a) \in \Delta_X$, $(a, a) \in \theta$. Hence $\Delta_X \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\Delta_X \subset \theta$. Let $a \in X$. By definition and assumption,
     $$\begin{aligned}(a, a) &\in \Delta_X \\ &\subset \theta.\end{aligned}$$
     Since $a \in X$ is arbitrary, we have that for each $a \in X$, $(a, a) \in \theta$. Hence θ is reflexive.
2.
   - $(\implies)$:
     Suppose that θ is symmetric. Let $(b, a) \in \theta^*$. By definition, $(a, b) \in \theta$. Since θ is symmetric, $(b, a) \in \theta$. Since $(b, a) \in \theta^*$ is arbitrary, we have that for each $(b, a) \in \theta^*$, $(b, a) \in \theta$. Hence $\theta^* \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\theta^* \subset \theta$. Let $a, b \in X$. Suppose that $(a, b) \in \theta$. By definition and assumption,
     $$\begin{aligned}(b, a) &\in \theta^* \\ &\subset \theta.\end{aligned}$$
     Since $a, b \in X$ with $(a, b) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ implies that $(b, a) \in \theta$. Hence θ is symmetric.
3.
   - $(\implies)$:
     Suppose that θ is transitive. Let $(a, c) \in \theta \circ \theta$. By definition, there exists $b \in X$ such that $(a, b) \in \theta$ and $(b, c) \in \theta$. Since θ is transitive, $(a, c) \in \theta$. Since $(a, c) \in \theta \circ \theta$ is arbitrary, we have that for each $(a, c) \in \theta \circ \theta$, $(a, c) \in \theta$. Hence $\theta \circ \theta \subset \theta$.
   - $(\impliedby)$:
     Suppose that $\theta \circ \theta \subset \theta$. Let $a, b, c \in X$. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. By definition and assumption,
     $$\begin{aligned}(a, c) &\in \theta \circ \theta \\ &\subset \theta.\end{aligned}$$
     Since $a, b, c \in X$ with $(a, b) \in \theta$ and $(b, c) \in \theta$ are arbitrary, we have that for each $a, b \in X$, $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$. Hence θ is transitive.

□

Definition 1.5.0.4. Let X be a set. Suppose that $X \neq \varnothing$. We define the **universal relation on** X, denoted $\nabla_X \in \mathcal{R}^2(X)$, by $\nabla_X := X^2$.

Exercise 1.5.0.5. Let X be a set. Suppose that $X \neq \varnothing$. Then $\nabla_X \in \mathrm{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \nabla_X$.
2. Suppose that $(a, b) \in \nabla_X$. Since $a, b \in X$, $(b, a) \in X^2$. Therefore $(a, b) \in \nabla_X$ implies that $(b, a) \in \nabla_X$.
3. Suppose that $(a, b), (b, c) \in \nabla_X$. Since $a, c \in X$, $(a, c) \in X^2$. Therefore $(a, b), (b, c) \in X^2$ implies that $(a, c) \in \nabla_X$.

□

Exercise 1.5.0.6. Let X be a set. Suppose that $X \neq \varnothing$. Then $\Delta_X \in \mathrm{Equiv}(X)$.

Proof. Let $a, b, c \in X$.

1. Since $a \in X$, $(a, a) \in \Delta_X$.
2. Suppose that $(a, b) \in \Delta_X$. Then $a = b$ and
$$\begin{aligned}(b, a) &= (a, a)\\ &\in \Delta_X.\end{aligned}$$
Therefore $(a, b) \in \Delta_X$ implies that $(b, a) \in \Delta_X$.
3. Suppose that $(a, b), (b, c) \in \Delta_X$. Then $a = b$ and $b = c$. Hence $a = c$ and
$$\begin{aligned}(a, c) &= (a, a)\\ &\in \Delta_X.\end{aligned}$$
Therefore $(a, b), (b, c) \in \Delta_X$ implies that $(a, c) \in \Delta_X$.

□

Exercise 1.5.0.7. Let X be a set. Suppose that $X \neq \varnothing$. Then for each $(\theta_\alpha)_{\alpha \in A} \subset \mathrm{Equiv}(X)$, $\bigcap_{\alpha \in A} \theta_\alpha \in \mathrm{Equiv}(X)$.

Proof. Let $(\theta_\alpha)_{\alpha \in A} \subset \mathrm{Equiv}(X)$. Set $\theta := \bigcap_{\alpha \in A} \theta_\alpha$. Let $a, b, c \in X$.

1. Let $\alpha \in A$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, $(a, a) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, a) \in \theta_\alpha$. Hence $(a, a) \in \theta$.
2. Suppose that $(a, b) \in \theta$. Let $\alpha \in A$. Then $(a, b) \in \theta_\alpha$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, we have that $(b, a) \in \theta_{a\backslash}$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(b, a) \in \theta_\alpha$. Hence $(b, a) \in \theta$. Therefore $(a, b) \in \theta$ implies that $(b, a) \in \theta$.
3. Suppose that $(a, b) \in \theta$ and $(b, c) \in \theta$. Let $\alpha \in A$. Then $(a, b), (b, c) \in \theta_\alpha$. Since $\theta_\alpha \in \mathrm{Equiv}(X)$, $(a, c) \in \theta_\alpha$. Since $\alpha \in A$ is arbitrary, we have that for each $\alpha \in A$, $(a, c) \in \theta_\alpha$. Hence $(a, c) \in \theta$. Therefore $(a, b) \in \theta$ and $(b, c) \in \theta$ implies that $(a, c) \in \theta$.

Hence $\theta \in \mathrm{Equiv}(X)$. □

Definition 1.5.0.8. Let X be a set. Suppose that $X \neq \varnothing$. Let $\nu \in \mathcal{R}^2(X)$.

- We define the **equivalence relations containing** ν, denoted $\mathcal{E}(\nu)$, by $\mathcal{E}(\nu) := \{\theta \in \mathrm{Equiv}(X) : \nu \subset \theta\}$.
- We define the **equivalence relation on** X **generated by** ν, denoted $\epsilon(\nu) \in \mathcal{R}^2(X)$ by $\epsilon(\nu) := \bigcap_{\theta \in \mathcal{E}(\nu)} \theta$.

Note 1.5.0.9.

- A previous ex implies that $\nabla_X \in \mathcal{E}(\nu)$.
- A previous ex (about intersections of equiv relats) implies that $\epsilon(\nu) \in \mathcal{E}(\nu)$.

Exercise 1.5.0.10. Let X be a set. Suppose that $X \neq \varnothing$. Let $\nu \in \mathcal{R}^2(X)$. Define $(\nu_n)_{n\in\mathbb{N}_0} \subset \mathcal{R}^2(X)$ by

$$\nu_n := \begin{cases} \Delta_X \cup \nu \cup \nu^*, & n = 0 \\ \nu_{n-1} \circ \nu_{n-1}, & n \geq 1. \end{cases}$$

Then

1. (a) for each $n \in \mathbb{N}$, ν_n is reflexive
   (b) for each $n \in \mathbb{N}$, ν_n is symmetric
   (c) for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$
2. $\epsilon(\nu) = \bigcup_{n\in\mathbb{N}_0} \nu_n$.

Proof.

1. (a)
   - **Base Case:**
     Since
     $$\begin{aligned} \Delta_X &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0, \end{aligned}$$
     a previous ex implies that ν_0 is reflexive.
   - **Induction Step:**
     Let $n \in \mathbb{N}_0$. Suppose that ν_n is reflexive. Then that same prev ex implies that $\Delta_X \subset \nu_n$. two prev ex in section on relations about composition with identity and subsets then imply that
     $$\begin{aligned} \Delta_X &= \Delta_X \circ \Delta_X \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}. \end{aligned}$$
     Another application of that prev ex implies that ν_{n+1} is reflexive.

   By induction, for each $n \in \mathbb{N}_0$, ν_n is reflexive.

   (b)
   - **Base Case:**
     a previous ex about converse relations and unions, double converse and since $\Delta_X \in \mathrm{Equiv}(X)$ and an ex about symmetric relations implies that
     $$\begin{aligned} \nu_0^* &= (\Delta_X \cup \nu \cup \nu^*)^* \\ &= \Delta_X^* \cup \nu^* \cup (\nu^*)^* \\ &= \Delta_X^* \cup \nu^* \cup \nu \\ &\subset \Delta_X \cup \nu^* \cup \nu \\ &= \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0. \end{aligned}$$
     that same previous ex about symmetric relations implies that ν_0 is symmetric.
   - **Induction Step:**
     Let $n \in \mathbb{N}_0$. Suppose that ν_n is symmetric. Then that same prev ex about symmetric and converse relations implies that $\nu_n^* \subset \nu_n$. Therefore prev exs imply
     $$\begin{aligned} \nu_{n+1}^* &= (\nu_n \circ \nu_n)^* \\ &= \nu_n^* \circ \nu_n^* \\ &\subset \nu_n \circ \nu_n. \end{aligned} \qquad = \nu_{n+1}.$$
     Another application of that prev ex implies that ν_{n+1} is symmetric.

   By induction, for each $n \in \mathbb{N}_0$, ν_n is symmetric.

(c) Let $n \in \mathbb{N}_0$. Since ν_n is reflexive, prev ex implies that $\Delta_X \subset \nu_n$. Therefore prev ex implies

$$\begin{aligned}\nu_n &= \Delta_X \circ \nu_n \\ &\subset \nu_n \circ \nu_n \\ &= \nu_{n+1}.\end{aligned}$$

Since $n \in \mathbb{N}_0$ is arbitrary, we have that for each $n \in \mathbb{N}_0$, $\nu_n \subset \nu_{n+1}$.

2. Set $\theta_0 := \bigcup_{n \in \mathbb{N}_0} \nu_n$.

- – **Base Case:**
  Since $\epsilon(\nu) \in \mathcal{E}(\nu)$, we have that $\epsilon(\nu) \in \mathrm{Equiv}(X)$ and $\nu \subset \epsilon(\nu)$. Since $\epsilon(\nu) \in \mathrm{Equiv}(X)$, we have that $\epsilon(\nu)$ is reflexive and symmetric. A ref ex then implies that $\Delta_X \subset \epsilon(\nu)$ and $\epsilon(\nu)^* \subset \epsilon(\nu)$. Therefore a prev ex about converse relations and subsets

$$\begin{aligned}\nu_0 &= \Delta_X \cup \nu \cup \nu^* \\ &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu)^* \\ &\subset \epsilon(\nu) \cup \epsilon(\nu) \cup \epsilon(\nu) \\ &= \epsilon(\nu).\end{aligned}$$

  – **Induction Step:**
  Let $n \in \mathbb{N}_0$. Suppose that $\nu_n \subset \epsilon(\nu)$. Then ref prev ex

$$\begin{aligned}\nu_{n+1} &= \nu_n \circ \nu_n \\ &\subset \epsilon(\nu) \circ \epsilon(\nu) \\ &= \epsilon(\nu).\end{aligned}$$

  By induction, for each $n \in \mathbb{N}_0$, $\nu_n \subset \epsilon(\nu)$. Therefore

$$\begin{aligned}\theta_0 &= \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &\subset \epsilon(\nu).\end{aligned}$$

- – Since ν_0 is reflexive, we have that

$$\begin{aligned}\Delta_X &\subset \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

  Hence θ_0 is reflexive.
  – Since for each $n \in \mathbb{N}_0$, ν_n is symmetric, we have that that ref exs for each $n \in \mathbb{N}_0$, $\nu_n^* \subset \nu_n$. Therefore

$$\begin{aligned}\theta_0^* &= (\bigcup_{n \in \mathbb{N}_0} \nu_n)^* \\ &= \bigcup_{n \in \mathbb{N}_0} \nu_n^* \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

  Hence θ_0 is symmetric.

– Let $a, b, c \in X$. Suppose that $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$. Then there exist $n_1, n_2 \in \mathbb{N}_0$ such that $(a, b) \in \nu_{n_1}$ and $(b, c) \in \nu_{n_2}$. Set $n_0 := \max\{n_1, n_2\}$. Since for each $n \in \mathbb{N}$, $\nu_n \subset \nu_{n+1}$, we have that

$$\begin{aligned}(a, b) &\in \nu_{n_1} \\ &\subset \nu_{n_0}.\end{aligned}$$

and

$$\begin{aligned}(b, c) &\in \nu_{n_2} \\ &\subset \nu_{n_0}.\end{aligned}$$

Since $(a, b), (b, c) \in \nu_{n_0}$, we have that

$$\begin{aligned}(a, c) &\in \nu_{n_0} \circ \nu_{n_0} \\ &= \nu_{n_0+1} \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0.\end{aligned}$$

Since $a, b, c \in X$ with $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ are arbitrary, we have that for each $a, b, c \in X$, $(a, b) \in \theta_0$ and $(b, c) \in \theta_0$ implies that $(a, c) \in \theta_0$. Hence θ_0 is transitive.

Since θ_0 is reflexive, symmetric and transitive, $\theta_0 \in \mathrm{Equiv}(X)$. Since

$$\begin{aligned}\nu &\subset \Delta_X \cup \nu \cup \nu^* \\ &= \nu_0 \\ &\subset \bigcup_{n \in \mathbb{N}_0} \nu_n \\ &= \theta_0,\end{aligned}$$

we have that $\theta_0 \in \mathcal{E}(\nu)$. Therefore

$$\begin{aligned}\epsilon(\nu) &= \bigcap_{\theta \in \mathcal{E}(\nu)} \theta \\ &\subset \theta_0.\end{aligned}$$

Since $\theta_0 \subset \epsilon(\nu)$ and $\epsilon(\nu) \subset \theta_0$, we have that $\theta_0 = \epsilon(\nu)$.

□

1.6 Products

Part I

Classical Physics

1.7 Basic Concepts (RENAME)

- Define length, time???, other units of measurement. Define position, velocity, acceleration, etc.
- Record experiment showing acceleration near us can be modeled as constant.
  - Use something simple with the same tools as Galileo. For example, release a block (try using oil or ice to reduce friction) from rest on a ramp and approximate velocity of a block at the bottom. Maybe you an use metronomes to binary search the amount of time the block travels a very small distance right before the bottom? Repeat for various heights, various weights. Repeat taking taking velocity measurements at various points along the ramp. Show that velocity at time t is proportional to the time elapsed. So acceleration is constant. Show the height of release (top of ramp) determines acceleration. Postulate that maximum acceleration happens at vertical ramp. Show that acceleration proportional to cosine of angle of incidence of ramp to horizontal (floor or table) and vertical is where cosine is 1 which is the maximum
  - Show that the velocity is proportional to the square root of height of release.
- Do experiment with pendulum (release at string aligned perpendicular to vertical), try to find a simple way to measure speed, show acceleration is constant and speed at bottom is proportional square root of string length (i.e. height of release)
- Do more general experiment where a block can slide on an organic surface (like a skateboarder in a skate park), so for example the graph of $z = -x^2 - y^2$ or other. Show that at different points of a level set, when released, the block slides in the direction of steepest descent (be precise in terms of paths on the surface, i.e. shortest paths on the surface i.e. geodesics)
- Talk about the idea of potential energy, and quantifying it in terms

1.8 Angular Momentum

- Come up with satisfactory model of angular momentum. Probably has somethign to do with areas, wedge products, etc
- Should be able to be generalized to manifolds
- Should be able to reproduce the electron like in https://quicycle.com/understanding-electrons/, see paper 'Is The Electron A Photon With Toroidal Topology?' by williamson, including on manifolds or spacetime
- Is there a good continuous model for densities on $\mathbb{R}^{2N}$ or more generally function spaces so that the angular momentum can be treated in a stat mech framework? https://physics.stackexchange.com/questions/650944/how-is-angular-momentum-defined-on-symplectic-space
-

Part II

Quantum Physics

Chapter 2

Spin

idea for getting around cloning theorem: While it is true that you cant copy the whole state of the system, this is also true for regular computations as well and various "classical categories" as well. Generally speaking a copy as a morphism copy : $X \to X \otimes X$ satisfying some property and we can use this to form the "tuple" of morphisms $(f, g) : X \to A \times B$ where $f : X \to A$ and $g : X \to B$ by $(f, g) := (f \otimes g) \circ \text{copy}$. We use (f, g) all the time in various contexts like in set theory or other math subjects, but the ability to form this morphism is equivalent to having a copy morphism (check this). However not all categories have such a morphism.

It is said that the NOR gate is functionally complete because you can form any logical operator $f : B^n \to B$ with $B = \{0, 1\}$ and $n \in \mathbb{N}_0$, but this is misleading. Using logic gates on different inputs, you cannot create a unary operator such as the identity or NOT gate just by using the NOR gate. This is because the arity of a composition of maps is bounded below by the lowest arity among the maps. So if you start with 2-ary maps, you can never get a 1-ary map. What is meant by this however, is that if you allow $f(g_1, \ldots, g_n)$, (i.e you consider the clone generated by the functions of interest, plus the porjections), then you can get all operators. Note that this is equivalent to copying because you form the tuple $(g_1, \ldots, g_n)$, so really you are not forming the NOT gate out of the NOR gate, but out of the NOR gate + copy.

(need to check this) The thing about quantum computing and the no cloning theorem is that the proof uses the assumption that you are clonging the entire information of the system. If we think about what is happening with the logic gates, the state of the system before and after processing thorugh a gate is described by the voltage and current, (at least in switch logic gates?). However for computation, we only use the voltage information. I think you can show that it is impossible to copy the voltage and current information simultaneously, so we basically copy the voltage information from before and after a COPY gate at the cost of splitting the current. Can you do something similar with Qbits? Can we descibe the state of the system using two pieces of info like spin and total spin squared S_x, S^2, they have a simultaneous basis since they commute, and compete similar to how voltage and current compete, maybe we can find a basis that we can copy only the spin info S at the cost of messing with the total spin info S^2?

Chapter 3

Quantization

Maybe change repo title to "into to quantum physics" instead of mechanics, that way we can cover field theory

- discuss how composite systems for particles are described by cartesian products. Then discuss how a particle at (x_0, p_0) can be described by a "field" (reall a prob dist) $\delta_{(x_0,p_0)}$. Then generally systems can be treated with prob dists and composite systems are described by tensor products, and that this generalizes the particle picture since $\delta_{(x_1,p_1)} \otimes \delta_{(x_2,p_2)} = \delta_{(x_1,p_1,x_2,p_2)}$.
- discuss Weil quantization, how as $\hbar \to 0$, we recover the posson bracket and commutative structure, discuss wigner transform of position and momentum functions giving position and momentum operators
- discuss rigged hilbert spaces to give meaning to "position basis", but treat as useful tool to get results like nonstandard analysis
- derive schrodinger equation from heisenberg picture
- free particle, harmonic oscillator, ladder operators, maybe hydrogen, do this in n-dimensions
- cover path integral, no complex measure exists, after rotation, prob measure on paths exists, then rotate back
- introduce field theory as first a theory of multiple particles and then as a continuum limit
- in the N particle case, the calculus of variations still works, remember, a quantum obervable operator $A \in HS(L^2(\mathbb{R}^N))$ corresponds to a classical observable function $f_A \in L^2(\mathbb{R}^{2N})$. We can write the lagrangian $L = \sum_j \mathcal{L}(x_1, \ldots, x_N, p_1, \ldots, p_N)$, which becomes $\int \mathcal{L}(\phi, \pi)$ in the continuum limit. Here $x_j = f_{X_j}$ and $p_j = f_{P_j}$ are the observeables corresponding to the position and momentum operators. We can then use calculus to find the $(x_j)_{j\in[N]}$, $(p_j)_{j\in[N]}$, i.e. ϕ and π which are stationary for $\mathcal{L}$. The question is then what quantization (wigner transform) means for these stationary ϕ and π. We can minimize the action $S[\phi] = \int \mathcal{L}(\phi, \pi)$ in $L^2(\mathbb{R}^{2N})$, but what does this correspond to in $HS(L^2(\mathbb{R}^N))$? does quantization preserve integration? i.e. $Q(\int \mathcal{L}((x_j), (p_j))) = \int Q(\mathcal{L}((x_j), (p_j))) = \mathcal{L}((X_j), (P_j))$?
- Since the isomorphism is between $HS(L^2\mathbb{R}^N)$ and $L^2(\mathbb{R}^{2N})$ with the star product, are we unable to guess what the true lagrangian is? For example, what if our system has classical lagrangian is xp. Since $x * p = p * x + O(\hbar)$, we wouldn't know if the true lagrangian was $x * p$, $p * x$, $(x * p + p * x)/2$ or something else. Can we distinguish this by experiment?
- Does the star product mess up the calculus of variations and do we still get the same euler lagrange equations with the star product? For lagrangians like the Klein-Gordon lagrangian with
- BIG GOAL: similar to how given a markov kernel $P^t(x, A) = P^t_x(A)$, we can define the operator Q_t in $C^\infty(M)$ by

$$Q_t f(x) := \int f(y)\, dP^t_x(y) = E[f(X_t)|X_0 = x]$$

  We can then find the infintesimal generator A such that

$$A := \frac{d}{dt}|_{t=0}(Q_t f)$$

which implies that $Q_t = e^{tA}$. Thinking in terms of Weyl quantization, we send observables $Q : L^2(\mathbb{R}^{2n}) \to HS(L^2(\mathbb{R}))$. The idea is that this is linear and in the state ρ, the expected value of observable $Q(f) = A_f$ given by $\langle A_f \rangle_\rho := \mathrm{tr}(A_f \rho)$ is linear in f. If this is continuous, then there should be a complex measure μ_ρ such that

$$\mathrm{tr}(A_f \rho) = \int f \, d\mu_\rho$$

. Similar to how brownian motion corresponds to the heat equation, can we define a quantum brownian motion to correspond to the free particle schrodinger equation using say the transition amplitude?

- BIG GOAL: Discuss envariance the derivation of borns rule, and statistical mechanics like microcanonical and canonical ensembles. For example, to derive the microcanonical distribution, a system $\mathcal{S}$ and environment $\mathcal{E}$ are in a joint state ψ which is maximally invariant. Then we can identify $\psi \in \mathcal{H}_\mathcal{S} \otimes \mathcal{H}_\mathcal{E}$ with the associated operator $A_\psi \in L(\mathcal{H}_\mathcal{E}, \mathcal{H}_\mathcal{S})$ and consider the singular value decomposition of A_ψ, (i.e. the Schmidt decomp of ψ) and write $\psi = \sum_k \alpha_k \psi_k^\mathcal{S} \otimes \psi_k^\mathcal{E}$. If ψ is invariant under all unitary operators $U_\mathcal{S}$ on $\mathcal{H}_\mathcal{S}$, we know that for each j, k, $|\alpha_j| = |\alpha_k|$ and we can take $(\psi_k^\mathcal{S})_k$ to be eigenvectors of $\mathcal{H}_\mathcal{S}$. If ψ is maximally energetically degenerate, so that for each k, $\mathcal{H}_\mathcal{S} \psi_k^\mathcal{S} = E \psi_k^\mathcal{S}$ (I think this is the case after measuring the energy and the composite system state collapses to one where the partial trace over the environment leaves us with a system state which is a linear combination degenerate eigenstates for example, but this should require an extra assumption like maximal envariance probably), then it follows that the probablilities of being in any one of the degenerate states are given by the uniform distribution over the number of degenerate states.

- to make sure I know relativity, take the continuum limit of couple harmonic oscillators to get a relativistic system (same lagrangian as klein-gordon) and then make the connection between this and relativity by describing signals in this system with relativity. What is an event, what is velocity, time, position, coordinates, etc. The idea is that since a relativistic equation describes the system, we should be able to use that language to talk about everything. However, this is also a classical system viewed from outside, so what is the connection. Can we do this in general and consider general relativity from a birds eye view in terms of a subsystem of a classical system which satisfies a relativistic equation?

3.1 Introduction

3.2

3.3 TODO

- Can we prepare various atoms in a certain state such that they bond in a way that yields interesting behavior? In other words, can we prepare atoms or subsystems in states, not necessarily all the same, such that we get interesting band structure or structure like cooper pairs, etc

Chapter 4

Quantum Fields

- discuss Schrodinger field as a continuum limit of a bunch of harmonic oscillators using creation annihilation operators and a lagrangian $L = \sum_{n \in \mathbb{Z}/N\mathbb{Z}} a_n^* a_n + a_{n+1}^* a_n + a_n^* a_{n+1}$ and explain how $a_{n+1}^* a_n + a_n^* a_{n+1}$ represents particles being destroyed at a location and created at an adjacent location. Show that this continuum limit is free particle. Then discuss more general lagrangians which allow for interactions like two-body interactions $V(n,m) a_n^* a_m^* a_m a_n$ and explain how we get interaction energy $V(n,m)$ if we have particles at sites n and m and so on with 3,4, ... particles see hitoshi murayama lectures

- Klein Gordon field as continuum limit of harmonic oscillators. Explore the case when the spring constants are not all constant, for instance maybe near $n = 0$, the spring constants get stronger/weaker. If we do the quantum field simulation done by ZAP physics, can we get represent the continuum limit using curvature? The idea here is that if the spring constant is stronger, the particle may pass by location $n = 0$ faster or slower and if we can get the particle to slow down enough as it approaches $n = 0$, can we get something like a schwarzchild radius to emerge?

- we can think of the classical fields as expected values of coherent states. Does this mean that the coherent states of an operator contain all information about operator?

4.1 Introduction

Appendix A

Summation

Definition A.0.0.1. Let $f : X \to [0, \infty)$, Then we define

$$\sum_{x \in X} f(x) := \sup_{\substack{F \subset X \\ F \text{ finite}}} \sum_{x \in F} f(x)$$

This definition coincides with the usual notion of summation when X is countable. For $f : X \to \mathbb{C}$, we can write $f = g + ih$ where $g, h : X \to \mathbb{R}$. If

$$\sum_{x \in X} |f(x)| < \infty,$$

then the same is true for g^+, g^-, h^+, h^-. In this case, we may define

$$\sum_{x \in X} f(x)$$

in the obvious way.

The following note justifies the notation $\sum_{x \in X} f(x)$ where $f : X \to \mathbb{C}$.

Note A.0.0.2. Let $f : X \to \mathbb{C}$ and $\alpha : X \to X$ a bijection. If $\sum_{x \in X} |f(x)| < \infty$, then $\sum_{x \in X} f(\alpha(x)) = \sum_{x \in X} f(x)$.

Appendix B

Asymptotic Notation

Definition B.0.0.1. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = o(g) \qquad \text{as } x \to x_0$$

if for each $\epsilon > 0$, there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U$,

$$\|f(x)\| \leq \epsilon \|g(x)\|$$

Exercise B.0.0.2. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. If there exists $U \in \mathcal{N}(x_0)$ such that for each $x \in U \setminus \{x_0\}$, $g(x) > 0$, then

$$f = o(g) \text{ as } x \to x_0 \quad \text{iff} \quad \lim_{x \to x_0} \frac{\|f(x)\|}{\|g(x)\|} = 0$$

Exercise B.0.0.3. Let X and Y a be normed vector spaces, $A \subset X$ open and $f : A \to Y$. Suppose that $0 \in A$. If $f(h) = o(\|h\|)$ as $h \to 0$, then for each $h \in X$, $f(th) = o(|t|)$ as $t \to 0$.

Proof. Suppose that $f(h) = o(\|h\|)$ as $h \to 0$. Let $h \in X$ and $\epsilon > 0$. Choose $\delta' > 0$ such that for each $h' \in B(0, \delta')$, $h' \in A$ and

$$\|f(h')\| \leq \frac{\epsilon}{\|h\| + 1} \|h'\|$$

Choose $\delta > 0$ such that for each $t \in B(0, \delta)$, $th \in B(0, \delta')$. Let $t \in B(0, \delta)$. Then

$$\begin{aligned} \|f(th)\| &\leq \frac{\epsilon}{\|h\| + 1} |t| \|h\| \\ &< \epsilon |t| \end{aligned}$$

So $f(th) = o(|t|)$ as $t \to 0$. $\square$

Definition B.0.0.4. Let X be a topological space, Y, Z be normed vector spaces, $f : X \to Y$, $g : X \to Z$ and $x_0 \in X \cup \{\infty\}$. Then we write

$$f = O(g) \qquad \text{as } x \to x_0$$

if there exists $U \in \mathcal{N}(x_0)$ and $M \geq 0$ such that for each $x \in U$,

$$\|f(x)\| \leq M \|g(x)\|$$

Appendix C

Categories

move to notation?

Definition C.0.0.1. We define the category of topological measure spaces, denoted $\mathbf{TopMsr}_+$, by

- $\mathrm{Obj}(\mathbf{TopMsr}_+) := \{(X, \mu) : X \in \mathrm{Obj}(\mathbf{Top}) \text{ and } \mu \in M(X)\}$
- $\mathrm{Hom}_{\mathbf{TopMsr}_+}((X, \mu), (Y, \nu)) := \mathrm{Hom}_{\mathbf{Top}}(X, Y) \cap \mathrm{Hom}_{\mathbf{Msr}_+}((X, \mathcal{B}(X), \mu), (Y, \mathcal{B}(Y), \nu))$

Appendix D

Vector Spaces

it might be better to cover some category theory and write everything in terms of $\mathrm{Hom}_{\mathbf{Vect}_{\mathbb{K}}}$ and $\mathrm{Obj}(\mathbf{Vect}_{\mathbb{K}})$

D.1 Introduction

Definition D.1.0.1. Let X be a set, $\mathbb{K}$ a field, $+ : X \times X \to X$ and $\cdot : \mathbb{K} \times X \to X$. Then $(X, +, \cdot)$ is said to be a **$\mathbb{K}$-vector space** if

1. $(X, +)$ is an abelian group
2.

Definition D.1.0.2. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$. Then $(E, +_E, \cdot_E)$ is said to be a subspace of X if

1. $+_E = +_X|_{E \times E}$
2. $\cdot_E = \cdot_X|_{\mathbb{K} \times E}$

Exercise D.1.0.3. Let $(X, +_X, \cdot_X)$ and $(E, +_E, \cdot_E)$ be vector spaces. Suppose that $E \subset X$.

Exercise D.1.0.4. Let $(X, +, \cdot)$ be a vector space and $E \subset X$. Then E is a subspace of X

Definition D.1.0.5. Let X be a vector space and $(E_j)_{j \in J}$ a collection of subspaces of X. Then $\bigcap_{j \in J} E_j$ is a subspace of X.

Proof. Set $E := \bigcap_{j \in J} E_j$. Let $x, y \in E$ and $\lambda \in \mathbb{K}$. Then for each $j \in J$, $x, y \in E_j$. Since for each $j \in J$, E_j is a subspace of X, we have that for each $j \in J$, $x + \lambda y \in E_j$. Thus $x + \lambda y \in E$. Since $x, y \in E$ and $\lambda \in \mathbb{K}$ are arbitrary, (cite exercise here) we have that E is a subspace of X. □

Definition D.1.0.6. Let X, Y be vector spaces and $T : X \to Y$. Then T is said to be **linear** if for each $x_1, x_2 \in X$ and $\lambda \in \Lambda$,

1. $T(x_1 + x_2) = T(x_1) + T(x_2)$,
2. $T(\lambda x_1) = \lambda T(x_1)$.

We define $L(X; Y) := \{T : X \to Y : \ T \text{ is linear}\}$.

Exercise D.1.0.7. Let X, Y be vector spaces and $T : X \to Y$. Then T is linear iff for each $x_1, x_2 \in X$ and $\lambda \in \Lambda$,

$$T(x_1 + \lambda x_2) = T(x_1) + \lambda T(x_2)$$

Proof. Clear. (add details) □

Definition D.1.0.8. define addition/scalar multiplication of linear maps

Exercise D.1.0.9. Let X, Y be vector spaces. Then $L(X;Y)$ is a $\mathbb{K}$-vector space.

Proof. Clear □

Definition D.1.0.10. Let X be a vector space over $\mathbb{K}$ and $T : X \to \mathbb{K}$. Then T is said to be a **linear functional on** X if T is linear. We define the **dual space of** X, denoted X^*, by $X^* := \{T : X \to \mathbb{K} : T \text{ is linear}\}$.

Exercise D.1.0.11. Let X be a vector space. Then X^* is a vector space.

Proof. Clear. □

D.2 Bases

Definition D.2.0.1. Let X be a vector space and $(e_\alpha)_{\alpha \in A} \subset X$. Then $(e_\alpha)_{\alpha \in A}$ is said to be

- **linearly independent** if for each $(\alpha_j)_{j=1}^n \subset A$, $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $\sum_{j=1}^n \lambda_j e_{\alpha_j} = 0$ implies that for each $j \in [n]$, $\lambda_j = 0$.
- a **Hamel basis for** X if $(e_\alpha)_{\alpha \in A}$ is linearly independent and $\operatorname{span}(e_\alpha)_{\alpha \in A} = X$.

Exercise D.2.0.2. every vector space has a Hamel basis

Proof. □

Exercise D.2.0.3.

Exercise D.2.0.4. Let X be a $\mathbb{K}$-vector space and $x \in X$. Then $x = 0$ iff for each $\phi \in X^*$, $\phi(x) = 0$.

Proof.

- $(\implies)$:
  Suppose that $x = 0$. Linearity implies that for each $\phi \in X^*$ $\phi(x) = 0$.
- $(\impliedby)$:
  Conversely, suppose that $x \neq 0$. Define $\epsilon_x : \operatorname{span}(x) \to \mathbb{K}$ by $\epsilon_x(\lambda x) := \lambda$. Let $u, v \in \operatorname{span}(x)$. Then there exists $\lambda_u, \lambda_v \in \mathbb{K}$ such that $u = \lambda_u x$ and $v = \lambda_v x$. Suppose that $u = v$. Then

  $$\begin{aligned}(\lambda_u - \lambda_v)x &= \lambda_u x - \lambda_v x \\ &= u - v \\ &= 0\end{aligned}$$

  Since $x \neq 0$, we have that $\lambda_u - \lambda_v = 0$ and therefore $\lambda_u = \lambda_v$. Hence

  $$\begin{aligned}\lambda_u &= \epsilon_x(u) \\ &= \epsilon_x(v) \\ &= \lambda_v.\end{aligned}$$

  Thus ϵ_x is well defined.

□

D.3 Multilinear Maps

Definition D.3.0.1. Let $X_1, \cdots, X_n, Y$ be vector spaces and $T : \prod_{j=1}^{n} X_j \to \mathbb{K}$. Then T is said to be **multilinear** if for each $j_0 \in [n]$ and $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$, $T(x_1, \ldots, x_{j_0-1}, \cdot, x_{j_0+1})$ is linear.

$$L^n(X_1, \ldots, X_n; Y) = \left\{ T : \prod_{j=1}^{n} X_j \to Y : T \text{ is multilinear} \right\}$$

If $X_1 = \cdots = X_n = X$, we write $L^n(X; Y)$ in place of $L^n(X, \ldots, X; Y)$.

Definition D.3.0.2. define addition and scalar mult of multilinear maps

Exercise D.3.0.3. Let $X_1, \cdots, X_n, Y$ be vector spaces. Then $L^n(X_1, \ldots, X_n; Y)$ is a $\mathbb{K}$-vector space.

Proof. content... □

Exercise D.3.0.4. Let $X_1, \cdots, X_n, Y, Z$ be $\mathbb{K}$-vector spaces, $\alpha \in L^n(X_1, \ldots, X_n; Y)$ and $\phi \in L^1(Y; Z)$. Then $\phi \circ \alpha \in L^n(X_1, \ldots, X_n; Z)$.

Proof. Let $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$ and $j_0 \in [n]$. Define $f : X_{j_0} \to Y$ by

$$f(a) := \alpha(x_1, \ldots, x_{j_0-1}, a, x_{j_0+1}, \ldots, x_n)$$

Since $\alpha \in L^n(X_1, \ldots, X_n; Y)$, f is linear. Since ϕ is linear, and $\phi \circ f$ is linear. Since $(x_j)_{j=1}^n \in \prod_{j=1}^{n} X_j$ and $j_0 \in [n]$ are arbitrary, we have that $\phi \circ \alpha \in L^n(X_1, \ldots, X_n; Y)$. □

D.4 Tensor Products

Definition D.4.0.1. Let X, Y and T be vector spaces over $\mathbb{K}$ and $\alpha \in L^2(X, Y; T)$. Then (T, α) is said to be a **tensor product of** X **and** Y if for each vector space Z and $\beta \in L^2(X, Y; Z)$, there exists a unique $\phi \in L^1(T; Z)$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & T \arrow[d, dashed, "\phi"] \\ & Z \end{tikzcd}$$

Exercise D.4.0.2. Let X, Y, S, T be vector spaces, $\alpha \in L^2(X, Y; S)$ and $\beta \in L^2(X, Y; T)$. Suppose that (S, α) and (T, β) are tensor products of X and Y. Then S and T are isomorphic.

Proof. Since (T, β) is a tensor product of X and Y, $\beta \in L^2(X, Y; T)$ there exists a unique $f \in L^1(T; T)$ such that $f\ circ\beta = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} & T \arrow[dd, dashed, "f"] \\ X \times Y \arrow[ur, "\beta"] \arrow[dr, "\beta"'] & \\ & T \end{tikzcd}$$

Since $\mathrm{id}_T \in L^1(T; T)$ and $\mathrm{id}_T \circ \beta = \beta$, we have that $f = \mathrm{id}_T$. Since (S, α) is a tensor product of X and Y, there exists a unique $\phi : S \to T$ such that $\phi \circ \alpha = \beta$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & S \arrow[d, dashed, "\phi"] \\ & T \end{tikzcd}$$

Similarly, since (T, β) is a tensor product of X and Y, there exists a unique $\psi : T \to S$ such that $\psi \circ \beta = \alpha$, i.e. the following diagram commutes:

$$\begin{tikzcd} X \times Y \arrow[r, "\beta"] \arrow[dr, "\alpha"'] & T \arrow[d, dashed, "\psi"] \\ & S \end{tikzcd}$$

Therefore

$$\begin{aligned} (\phi \circ \psi) \circ \beta &= \phi \circ (\psi \circ \beta) \\ &= \phi \circ \alpha \\ &= \beta, \end{aligned}$$

i.e. the following diagram commutes:

$$\begin{tikzcd} & T \arrow[d, dashed, "\psi"] & \\ X \times Y \arrow[ur, "\beta"] \arrow[r, "\alpha"] \arrow[dr, "\beta"'] & S \arrow[d, dashed, "\phi"] & \\ & T & \end{tikzcd} \implies \begin{tikzcd} & T \arrow[dd, dashed, "\phi\circ\psi"] \\ X \times Y \arrow[ur, "\beta"] \arrow[dr, "\beta"'] & \\ & T \end{tikzcd}$$

By uniqueness of $f \in L^1(T; T)$, we have that

$$\begin{aligned} \mathrm{id}_T &= f \\ &= \phi \circ \psi \end{aligned}$$

A similar argument implies that $\psi \circ \phi = \mathrm{id}_S$. Hence ϕ and ψ are isomorphisms with $\phi^{-1} = \psi$. Hence S and T are isomorphic. $\square$

Definition D.4.0.3. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. We define $x \otimes y : X^* \times Y^* \to \mathbb{K}$ by $x \otimes y(\phi, \psi) := \phi(x)\psi(y)$.

Exercise D.4.0.4. Let X, Y be vector spaces, $x \in X$ and $y \in Y$. Then $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$.

Proof. Let $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$. Then

$$\begin{aligned} x \otimes y(\phi_1 + \lambda\phi_2, \psi) &= [\phi_1 + \lambda\phi_2(x)]\psi(y) \\ &= \phi_1(x)\psi(y) + \lambda\phi_2(x)\psi(y) \\ &= x \otimes y(\phi_1, \psi) + \lambda x \otimes y(\phi_2, \psi) \end{aligned}$$

Since $\phi_1, \phi_2 \in X^*$, $\psi \in Y^*$ and $\lambda \in \mathbb{K}$ are arbitrary, we have that for each $\psi \in Y^*$, $x \otimes y(\cdot, \psi)$ is linear. Similarly for each $\phi \in X^*$, $x \otimes y(\phi, \cdot)$ is linear. Hence $x \otimes y$ is bilinear and $x \otimes y \in L^2(X^*, Y^*; \mathbb{K})$. □

Definition D.4.0.5. Let X, Y be vector spaces. We define

- the **tensor product of X and Y**, denoted $X \otimes Y \subset L^2(X^*, Y^*; \mathbb{K})$, by
$$X \otimes Y := \operatorname{span}(x \otimes y : x \in X \text{ and } y \in Y),$$
- the **tensor map**, denoted $\otimes : X \times Y \to X \otimes Y$, by $\otimes(x, y) := x \otimes y$.

Exercise D.4.0.6. Let X, Y be vector spaces, $(x_j)_{j=1}^n \subset X$ and $(y_j)_{j=1}^n \subset Y$. The following are equivalent:

1. $\sum_{j=1}^n x_j \otimes y_j = 0$
2. for each $\phi \in X^*$ and $\psi \in Y^*$, $\sum_{j=1}^n \phi(x_j)\psi(y_j) = 0$
3. for each $\phi \in X^*$, $\sum_{j=1}^n \phi(x_j)y_j = 0$
4. for each $\psi \in Y^*$, $\sum_{j=1}^n \psi(y_j)x_j = 0$

Proof.

1. $(1) \implies (2)$:
   Suppose that $\sum_{j=1}^n x_j \otimes y_j = 0$. Let $\phi \in X^*$ and $\psi \in Y^*$. Then
$$\begin{aligned} \sum_{j=1}^n \phi(x_j)\psi(y_j) &= \phi\left(\sum_{j=1}^n \psi(y_j)x_j\right) \\ &= \end{aligned}$$
2.
3.

□

Exercise D.4.0.7. Let X, Y be vector spaces. Then $(X \otimes Y, \otimes)$ is a tensor product of X and Y.

Proof. Let Z be a vector space and $\alpha \in L^2(X, Y; Z)$. Define $\phi : X \otimes Y \to Z$ by $\phi\left(\sum_{j=1}^n \lambda_j x_j \otimes y_j\right) := \sum_{j=1}^n \lambda_j \alpha(x_j, y_j)$.

- **(well defined):**
  Let $u \in X \otimes Y$. Then there exist $(\lambda_j)_{j=1}^n \subset \mathbb{K}$, $(x_j)_{j=1}^n \subset X$, $(y_j)_{j=1}^n \subset Y$ such that $u = \sum_{j=1}^n \lambda_j x_j \otimes y_j$. Suppose that $u = 0$. Let $\phi \in Z^*$. Then $\phi \circ \alpha \in L^2(X, Y; Z)$.

□

Bibliography

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